An Autonomous Mobile Robot for Line Following, Wall Detection and Ball Carriage

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Report type	Technical report, team design project
Team and role	Team of 3. Individual contribution: chassis design, sensor electronics and control-software tuning
Status	Built and tested on the course; 3 of 4 objectives completed
Tools	Arduino Nano ESP32, L298N, HC-SR04, SolidWorks, embedded C (Arduino framework)
Date	2026

Abstract. A mobile robot was designed to complete a multi-stage course within a 220 mm cube: follow a curved black line, stop within 300 mm of a wall, reverse into a parking area, and carry a ball throughout. The design combines a laser-cut acrylic chassis with a low centre of mass, two light-dependent-resistor (LDR) line sensors, an ultrasonic range-finder, and a proportional differential-steering law with gain k_p = 0.4. An encoder-based inner speed loop was implemented but disabled after it interacted with the outer line-following loop and caused oscillation, which is consistent with insufficient bandwidth separation between nested loops. Line following, the wall stop and ball carriage were completed; parking, which depended on encoder odometry, was not.

Index terms: mechatronics, line following, proportional control, differential steering, ultrasonic ranging, cascaded control.

1. Problem Definition and Requirements

Table 1. Course requirements.

Requirement	Target
R1 Line following	Follow a curved black line on a white surface
R2 Wall stop	Stop within 300 mm of a wall
R3 Parking	Reverse into a marked parking area
R4 Ball carriage	Carry a ball for the full course
R5 Envelope	Fit within a 220 mm cube

2. System Architecture

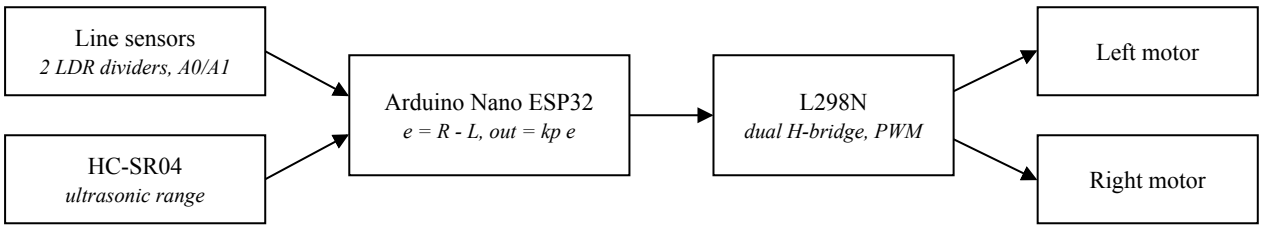


Figure 1. Sense, compute and actuate structure. Two LDR voltage dividers and an ultrasonic sensor feed the microcontroller, which drives two rear DC gear motors through an L298N H-bridge (ENA/ENB PWM and IN1 to IN4 direction). A 7.4 V pack supplies the motor rail; logic runs at 5 V.

3. Mechanical Design

A 200 x 150 mm chassis plate was designed in SolidWorks and laser-cut from 3 mm acrylic. The drive motors are mounted symmetrically at the rear for differential steering, with a 3D-printed roller-ball caster at the front. The electronics are placed centrally over the driven axle to increase traction, and the ball cup sits near the centre-rear turning axis to reduce the lateral acceleration on the ball during turns.

4. Sensing

Two LDRs with illuminating LEDs are mounted on a 3D-printed plate at ±35 mm from the centreline. Reflected light falls over the black line, increasing LDR resistance; each LDR forms a voltage divider read on an analogue input. The difference between the two readings is a signed error whose sign gives the direction of deviation. A 40 mm foam hood shields the sensors from ambient light. The HC-SR04 measures wall distance by ultrasonic time of flight (approximately 58 µs per cm of range).

5. Control Law

$$e = R_{LDR} - L_{LDR}, \quad out = k_p \cdot e, \quad left = 100 + out, \quad right = 100 - out \tag{1}$$

The base PWM of 100 (of 255) leaves headroom for correction in both directions. The gain was tuned empirically: below $k_p = 0.3$ the robot lost the line on tight curves, and above 0.6 it over-corrected and weaved; $k_p = 0.4$ was selected. Below a 300 mm range reading both motors are commanded to zero.

6. Design Decisions

- A three-LDR sensor was rejected because its algorithm required calibration time not available in the schedule; the two-sensor error was sufficient.
- Underneath and rear battery positions were rejected because of poor packaging and a centre-of-mass shift that reduced traction.
- Additional ball-retention features (side supports, a pendulum) were tested and removed once the cup geometry alone retained the ball.

7. Cascaded-Control Failure Analysis

Attempted design. Each motor carried a quadrature encoder intended to close an inner speed loop, equalising wheel speeds against motor tolerance and battery sag, beneath the outer LDR line-following loop.

Observation. With both loops active the robot oscillated and line following degraded relative to the proportional controller alone.

Diagnosis. A brushed DC motor is approximately first order from voltage to speed, with mechanical time constant $\tau_m = J \cdot R / (K_t \cdot K_e)$ [2]. A nested loop behaves as intended only when the inner loop is substantially faster than the outer loop, conventionally by a factor of five to ten in bandwidth:

$$\omega_{inner} \geq (5 \text{ to } 10) \cdot \omega_{outer} \tag{2}$$

The two loops ran at comparable rates, so each acted as a lagging dynamic element for the other, reducing phase margin. This explanation is consistent with the observed behaviour; loop bandwidths were not measured.

Decision. The encoder loop was disabled and the stable proportional controller retained. Parking depended on encoder distance counting and could not be completed.

8. Results

Table 2. Requirement verification (hardware test on the course).

Requirement	Result	Status
R1 Line following	Completed on the course	Met
R2 Wall stop	Stopped below the 300 mm threshold	Met
R3 Parking	Not completed (encoder loop disabled)	Not met
R4 Ball carriage	Ball retained throughout	Met
R5 Envelope	200 x 150 mm chassis within 220 mm cube	Met

9. Limitations

- The gain was tuned from qualitative observation; lateral error, completion time and line-loss events were not measured as functions of k_p.
- Results are reported as completed or not completed; repeat-trial statistics were not recorded.
- The bandwidth-separation diagnosis is inferred from behaviour rather than from measured loop responses.

10. Conclusion and Future Work

The robot met four of five requirements. The principal technical finding is the interaction between the nested speed and position loops. Future work: redesign the inner speed loop to run five to ten times faster than the line loop, with a filtered velocity estimate, to restore odometry for parking; add a derivative term to the line loop to reduce weave on tight curves; and characterise tracking error against k_p quantitatively.

Appendix A. Firmware

```
const int LDR_L = A0, LDR_R = A1;          // line-sensor dividers
const int ENA = 5, IN1 = 6, IN2 = 7;       // left motor
const int ENB = 9, IN3 = 8, IN4 = 10;      // right motor
const int TRIG = 2, ECHO = 3;              // HC-SR04
const int BASE = 100; const float KP = 0.4; const int WALL_MM = 300;

long readRangeMM() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10); digitalWrite(TRIG, LOW);
  long us = pulseIn(ECHO, HIGH, 30000);     // 30 ms timeout
  return us * 10L / 58L;                    // approx. 58 us per cm
}
void drive(int l, int r) { analogWrite(ENA, constrain(l, 0, 255)); analogWrite(ENB, constrain(r, 0, 255)); }

void loop() {
  if (readRangeMM() < WALL_MM) { drive(0, 0); return; }      // wall stop
  int e = analogRead(LDR_R) - analogRead(LDR_L);             // signed error
  int out = (int)(KP * e);
  drive(BASE + out, BASE - out);                             // differential steering
}
```

Listing A1. Control loop (pin setup omitted). The encoder loop is not included.

References

[1] R. Siegwart, I. R. Nourbakhsh and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed. MIT Press, 2011.

[2] K. Ogata, *Modern Control Engineering*, 5th ed. Pearson, 2010.

[3] W. Bolton, *Mechatronics*, 7th ed. Pearson, 2018.

[4] STMicroelectronics, "L298: Dual full-bridge driver," datasheet.

[5] "HC-SR04 ultrasonic ranging module," manufacturer datasheet.